Activation energy and the Gamow window

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + \mathrm{n} + 17,6\ \mathrm{MeV}$$

Q>0 !

- Exothermic although does not happen spontaneously!

Activation energy and the Gamow window

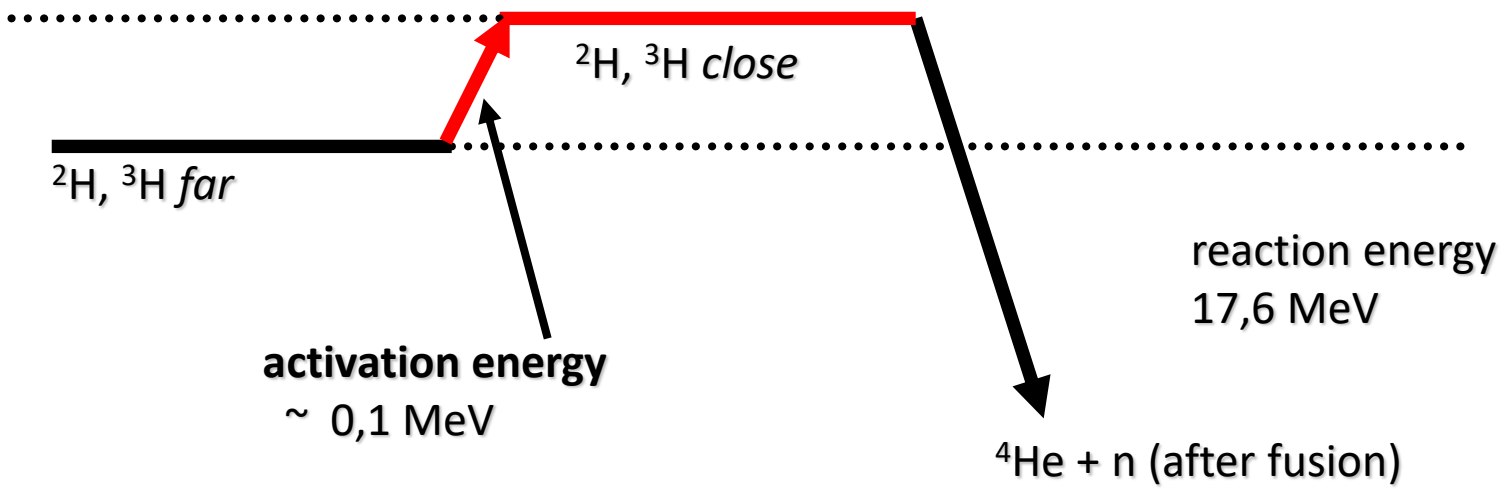
$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + \mathrm{n} + 17,6\ \mathrm{MeV}$$

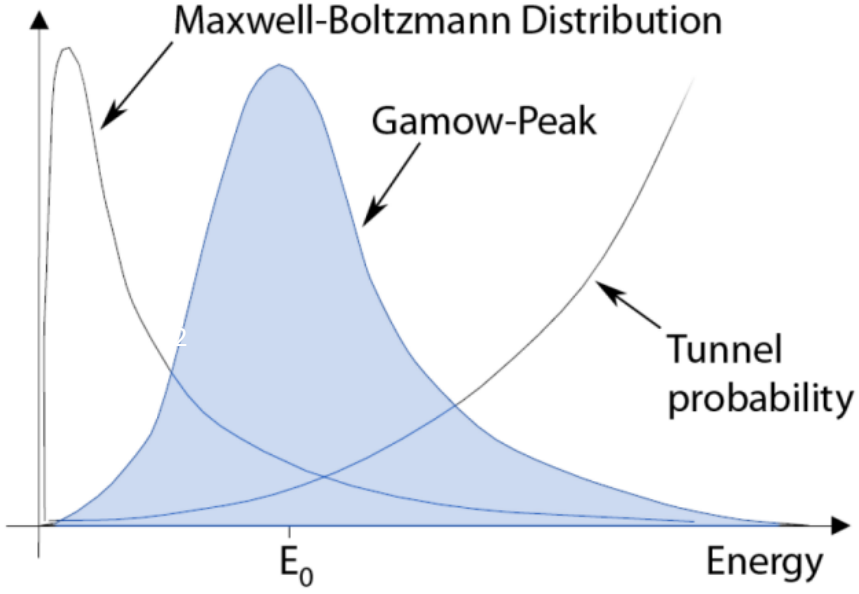
Q>0 !

- Exothermic although does not happen spontaneously!
- Reason: Coulomb repulsion
- Extra kinetic energy is needed. But remember ⟶ tuenneling!

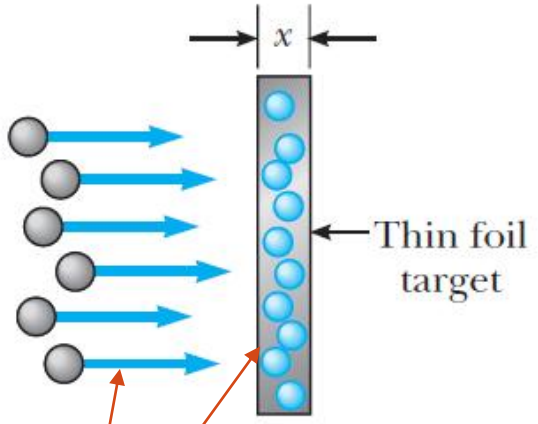
$$P_{\mathrm{MB}}(E) \propto e^{-m\langle v^2\rangle/2k_{\mathrm{B}}T} = e^{-E/k_{\mathrm{B}}T}$$

$$P_{\mathrm{G}}(E) = e^{-\sqrt{E_{\mathrm{G}}/E}}$$





The cross section and reaction rate



$$R = (\Phi A)\left(\frac{\rho A \Delta x \sigma}{A}\right) = N\Phi\sigma$$

$$\sigma = \frac{R}{N \cdot \Phi}$$

Nuclear reactions are fundamentally quantum mechanical processes

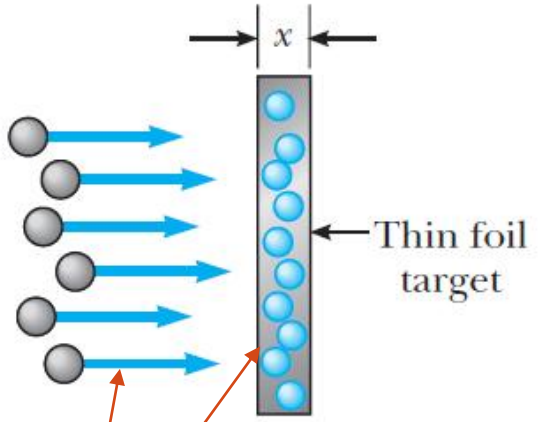
microscopic cross section: **σ**

- Number of target atoms
- Reaction rate: [R]=1/s
- Incoming flux(density) [Φ]= 1/(cm²s)

- The probability that an incoming particle „hits" an atom: $\frac{N\sigma}{A}$
- The expected number of reactions taking place over unit time: $R = \frac{(\Phi A)N\sigma}{A}$

Determines the probability of a certain reaction. [σ]=cm²; 10^{-24} cm²= 1 barn

The cross section and reaction rate



Nuclear reactions are fundamentally quantum mechanical processes

„Measurement" definition: $\sigma = \dfrac{R}{N \cdot \Phi}$ (R: final state events; N·Φ: initial state events)

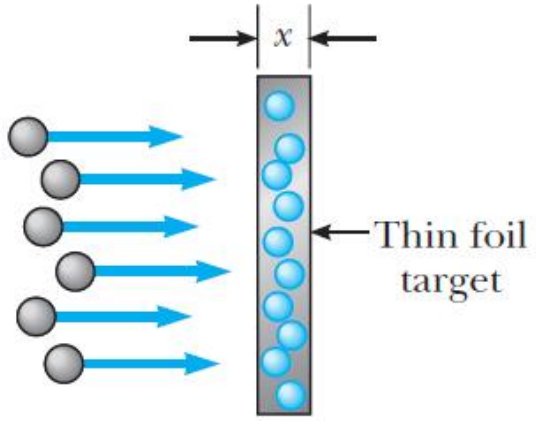
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The cross section and reaction rate



$$R = (\Phi A)\left(\frac{\rho A \Delta x \sigma}{A}\right) = N\Phi\sigma$$

$$\sigma = \frac{R}{N \cdot \Phi}$$

$$n + {}^{235}_{92}U \rightarrow \begin{cases} {}^{236}_{92}U + \gamma & \sigma_c \rightarrow \text{absorption: capture} \\ {}^{235}_{92}U + n' + \gamma & \sigma_{is} \rightarrow \text{scattering} \\ fission & \sigma_f \rightarrow \text{absorption: fission} \end{cases}$$

If the particles inducing the different reactions are the same, N and Φ do not change.

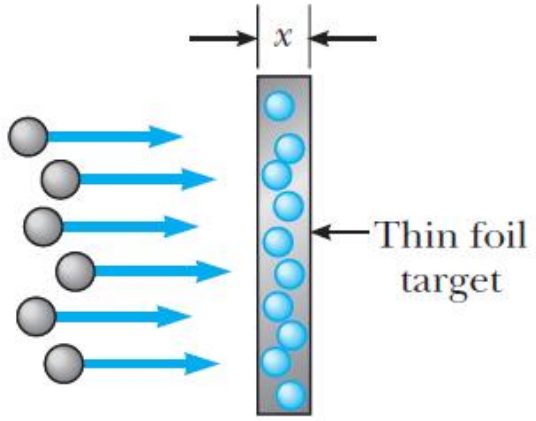
$$R_{total} = R_{scattering} + R_{absorption} = N\Phi\sigma_t$$

$$\frac{R_t}{N\Phi} = \frac{R_s}{N\Phi} + \frac{R_a}{N\Phi}$$

$$\sigma_t = \sigma_s + \sigma_a$$

For the independent processes the probabilities are additive

The cross section and reaction rate



$$R = (\Phi A)\left(\frac{\rho A \Delta x \sigma}{A}\right) = N\Phi\sigma$$

$$\sigma = \frac{R}{N \cdot \Phi}$$

$$n + {}^{235}_{92}U \rightarrow \begin{cases} {}^{236}_{92}U + \gamma & \sigma_c \rightarrow \text{absorption: capture} \\ {}^{235}_{92}U + n' + \gamma & \sigma_{is} \rightarrow \text{scattering} \\ fission & \sigma_f \rightarrow \text{absorption: fission} \end{cases}$$

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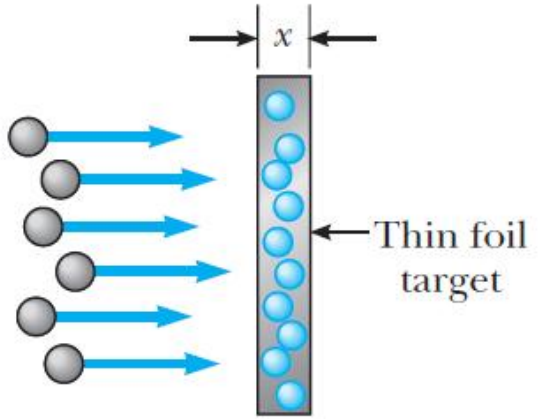
$$\sigma_t = \sigma_s + \sigma_a$$

For the independent processes the probabilities are additive

$$\sigma(E) = \frac{R}{\Phi(E)N}$$

Generally the cross section depends on the energy of the incident particle

The cross section and reaction rate



$$R = (\Phi A)\left(\frac{\rho A \Delta x \sigma}{A}\right) = N\Phi\sigma$$

$$\sigma = \frac{R}{N \cdot \Phi}$$

How the beam intensity decreases after passing through a target?

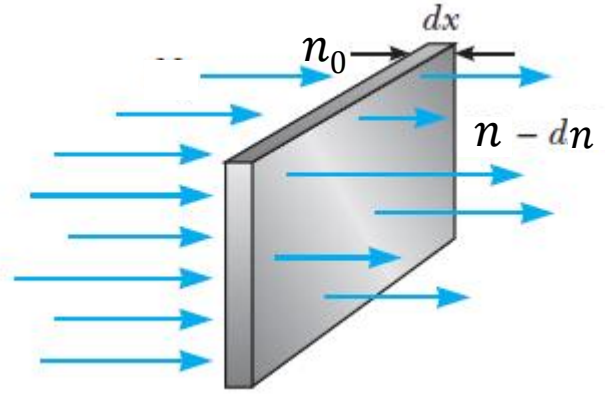
Number of particles consumed = reaction rate

$$\frac{-dn}{\Phi A} = \frac{\sigma\rho A dx}{A} = \sigma\rho dx$$

Number of incident particles: n= ΦA $[1/s]$

$$\Sigma = \rho\sigma \qquad [cm^{-1}]$$

Macroscopic property:
Macroscopic cross section



$$\int_{n_0}^{n} \frac{dn}{n} = -\sigma\rho \int_{0}^{x} dx$$

$$n(x) = n_0 e^{-\sigma\rho x}$$

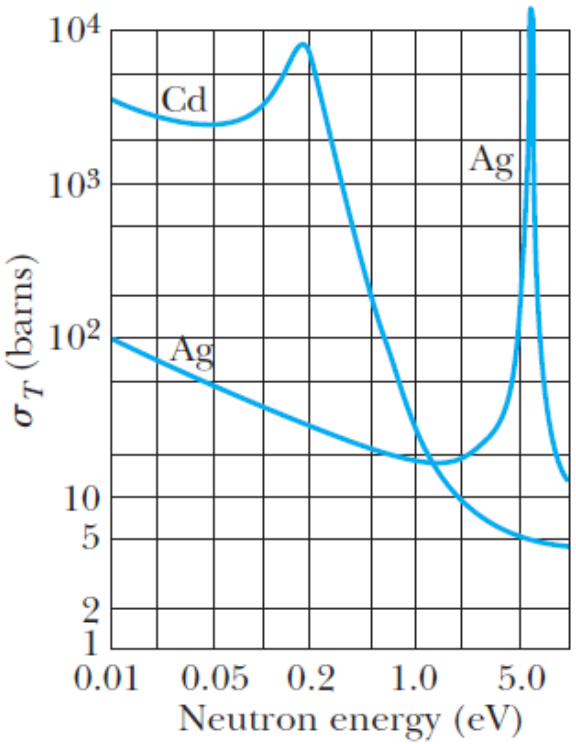
$$\Phi(x) = \Phi_0 e^{-\Sigma x}$$

Neutron-activated nuclear reactions

General properties:

- Absence of Coulomb-interaction
  - Moderate reactivity with matter
  - Similar statistical properties to photons
  - After many scattering the neutron gas thermalizes

$$K_{av} = \frac{3}{2} k_B T \approx 0.04 \text{ eV}$$



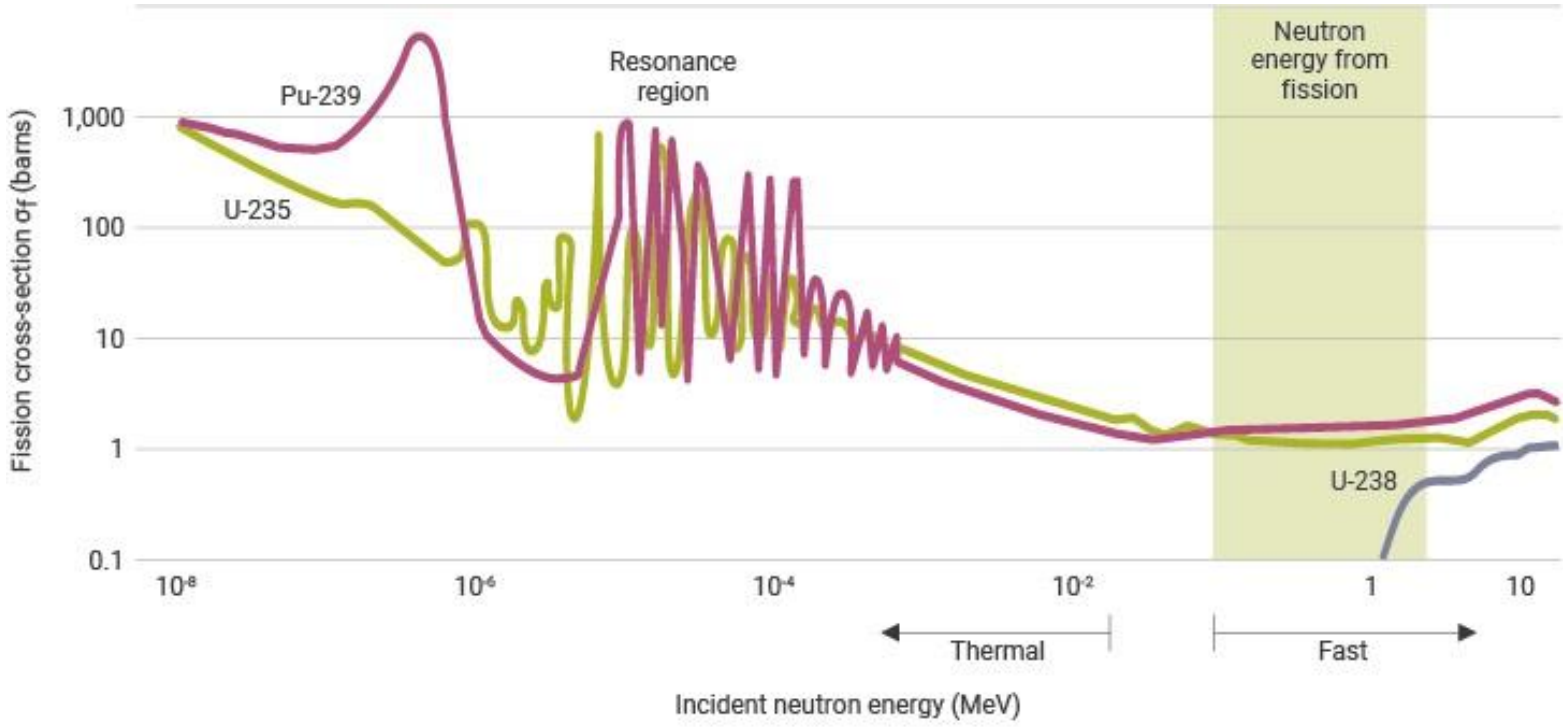
Typical and relevant reactions:

- Neutron absorption reactions: (n,γ) or fission
  - **1/v cross section** for not too high energies: the longer the neutron is around the nucleus the more probable it gets captured

$$n + {}^{113}Cd \longrightarrow {}^{114}Cd + \gamma \qquad (\sigma = 10^4 \text{ b})$$

- Elastic/inelastic scattering
  - Max energy transfer for collision with protons (with materials abundant in H) and for **high-energy neutrons** (exept for resonances)

The fission cross section for U and Pu

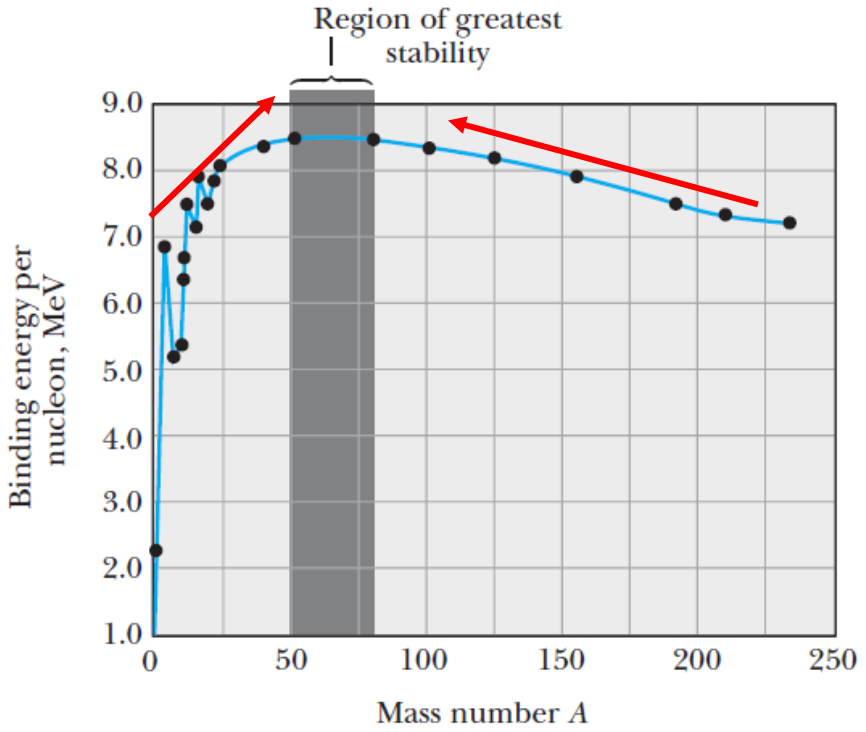


Nuclear fission and fission reactors

ENERGY RELEASED, DEFORMATION, CROSS SECTION, PROMPT AND DELAYED NEUTRONS, CHAIN REACTION, THE DESTINY OF NEUTRONS, MODERATORS

Nuclear energy release: fission

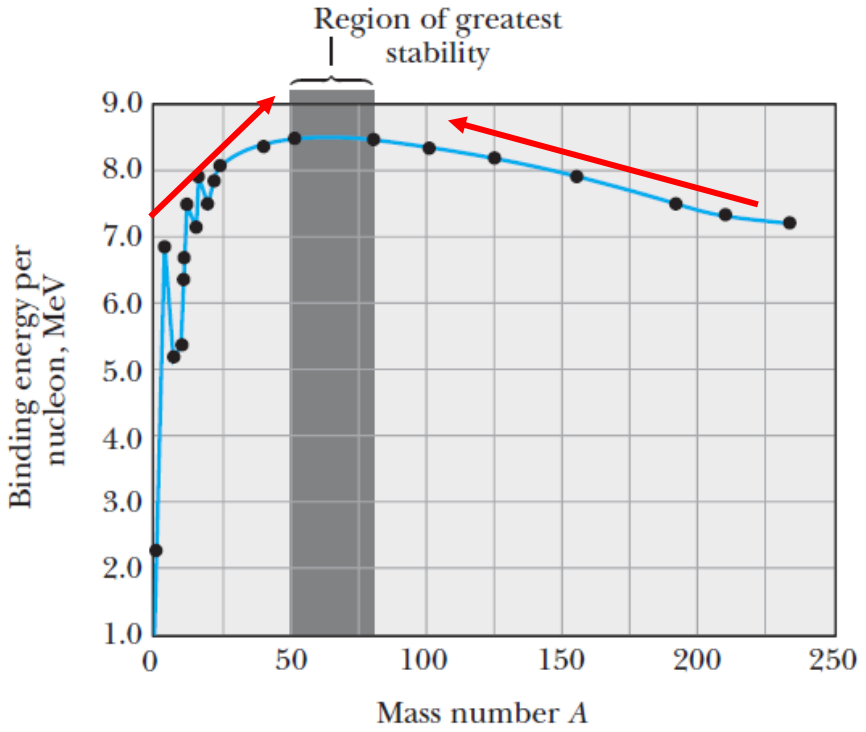
- **Fusion**
- Large energy gain /nucleon (2-5 MeV)
- Few nucleons
- Energy gain: **2-18 MeV**



- **Fission**
- Smaller energy gain /nucleon (around 0.8 MeV)
- Many nucleons!
- Larger total energy gain: **200 MeV**

Nuclear energy release: fission

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- Many nucleons!
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Fission reaction

Reaction of fission:

$^{235}U + n \longrightarrow$ **Fission product$_1$ + Fission product$_2$ +** 2,4 n

Fission reaction

Reaction of fission:

$^{235}U + n \longrightarrow$ **Fission product$_1$ + Fission product$_2$ +** 2,4 n

Induced fission

Different reaction channels

Expected number of prompt neutrons

Fission is a probabilistic process!

Fission product yield

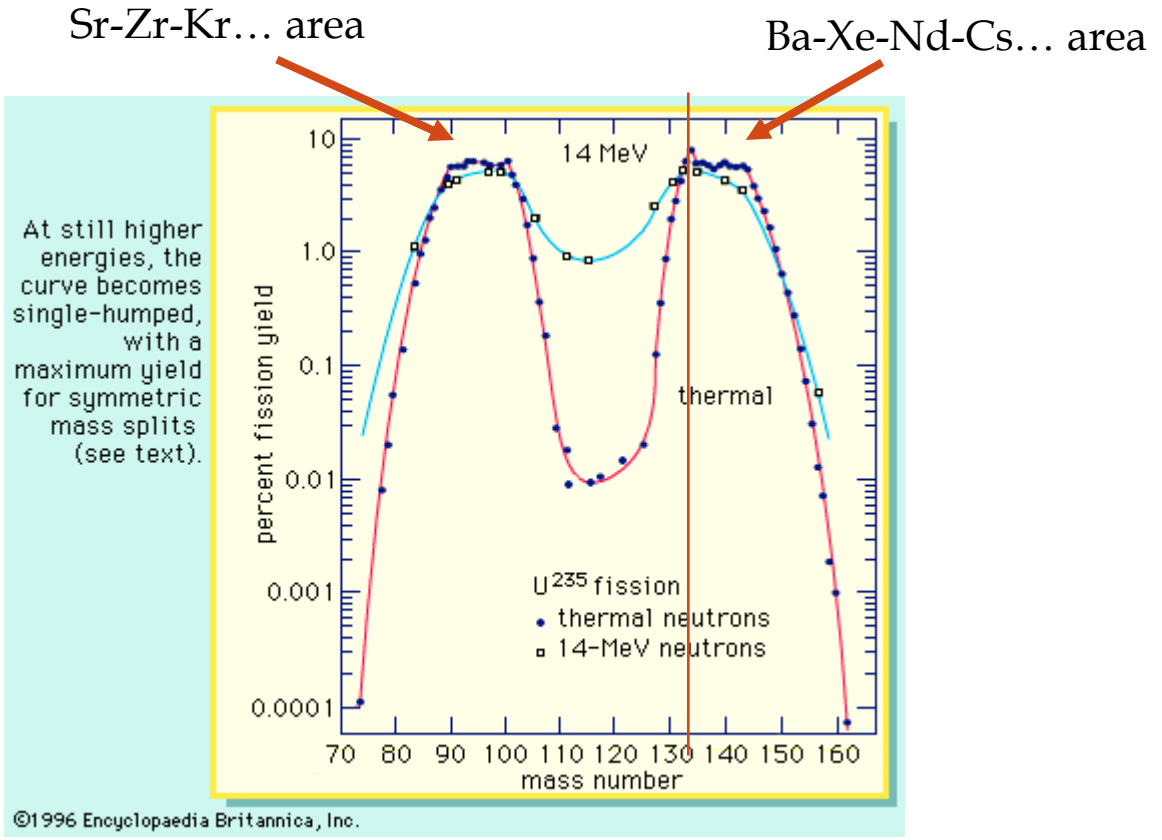
Investigate many fission events

- **Two maximums**
  - **Asymmetric fission**
  - Curve not smooth: even-odd effect
  - higher mass-number peak is stable

- The *higher the mass number* of the nucleus undergoing fission or the *higher the energy of the neutron* inducing the fission the more likely a *symmetric fission* is

$$^{236}_{92}U \rightarrow ^{118}_{46}Pd + ^{118}_{46}Pd$$

- For *spontaneous fission* (typical for extremely large mass-number nuclei) symmetric fission product yield



Fission product yield

Investigate many fission events

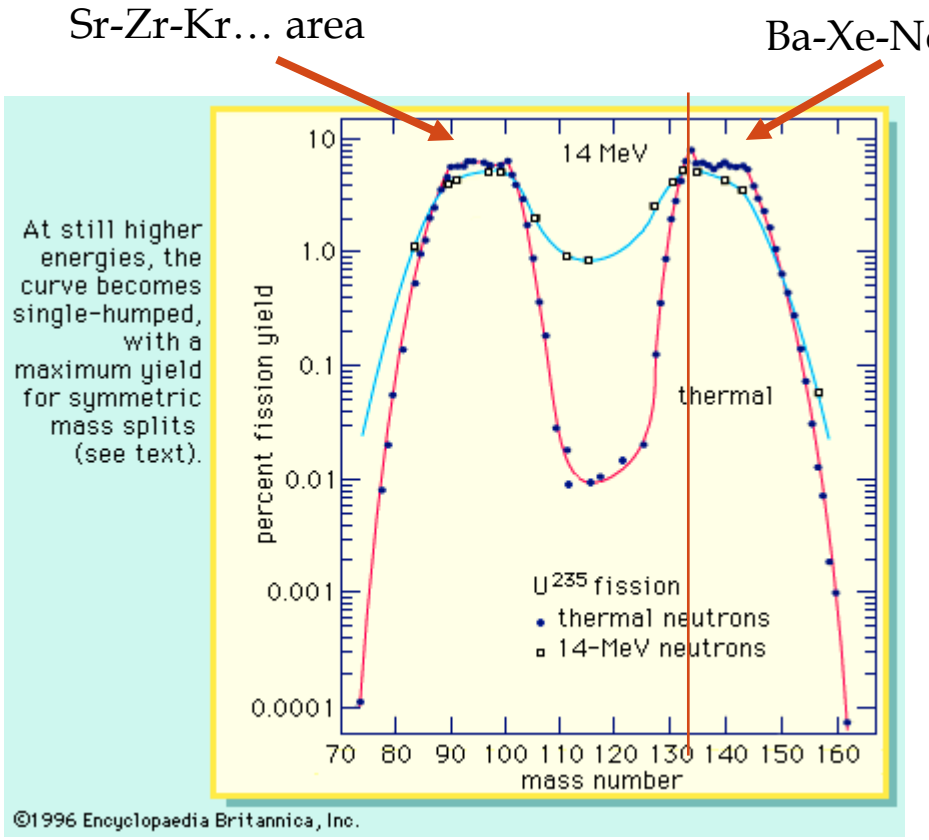
- **Two maximums**
  - **Asymmetric fission**
  - Curve not smooth: even-odd effect
  - higher mass-number peak is stable

asymmetric fission:

$$^{236}\mathrm{U} \longrightarrow {}^{90}\mathrm{Kr} + {}^{143}\mathrm{Ba} + 3\mathrm{n}$$

symmetric fission:

$$^{236}_{92}U \rightarrow {}^{118}_{46}Pd + {}^{118}_{46}Pd$$

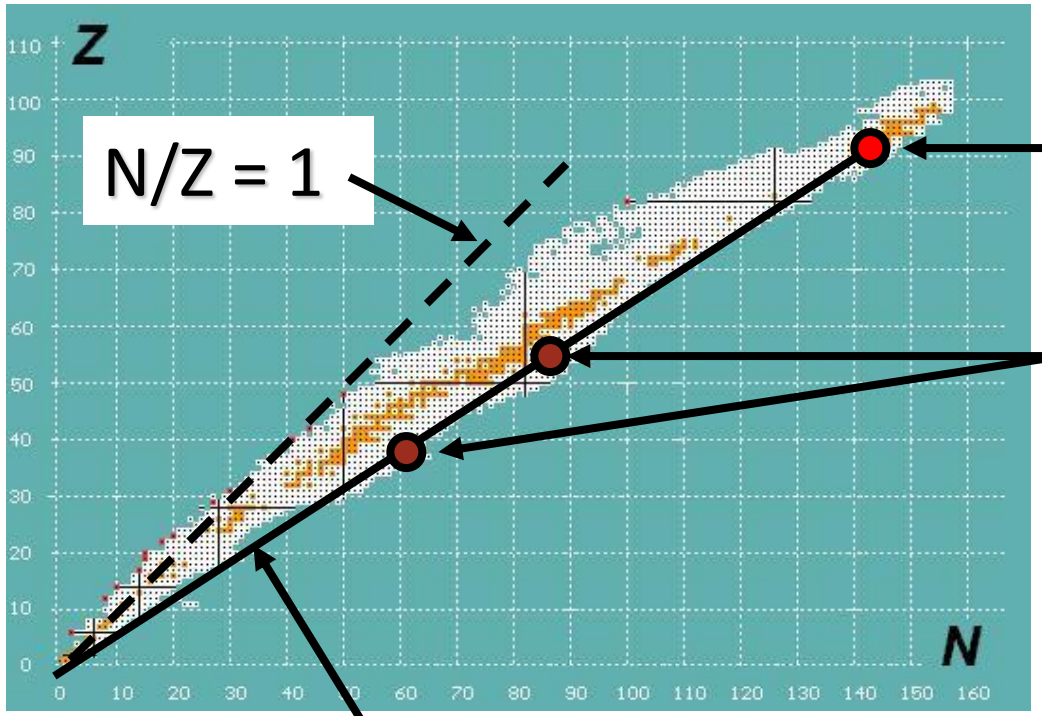


Nuclear fission: neutron production

$^{236}U \longrightarrow {}^{90}Kr + {}^{143}Ba + 3n$

$^{236}_{92}U_{144}$

$N/Z = 144/92 = 1,56$

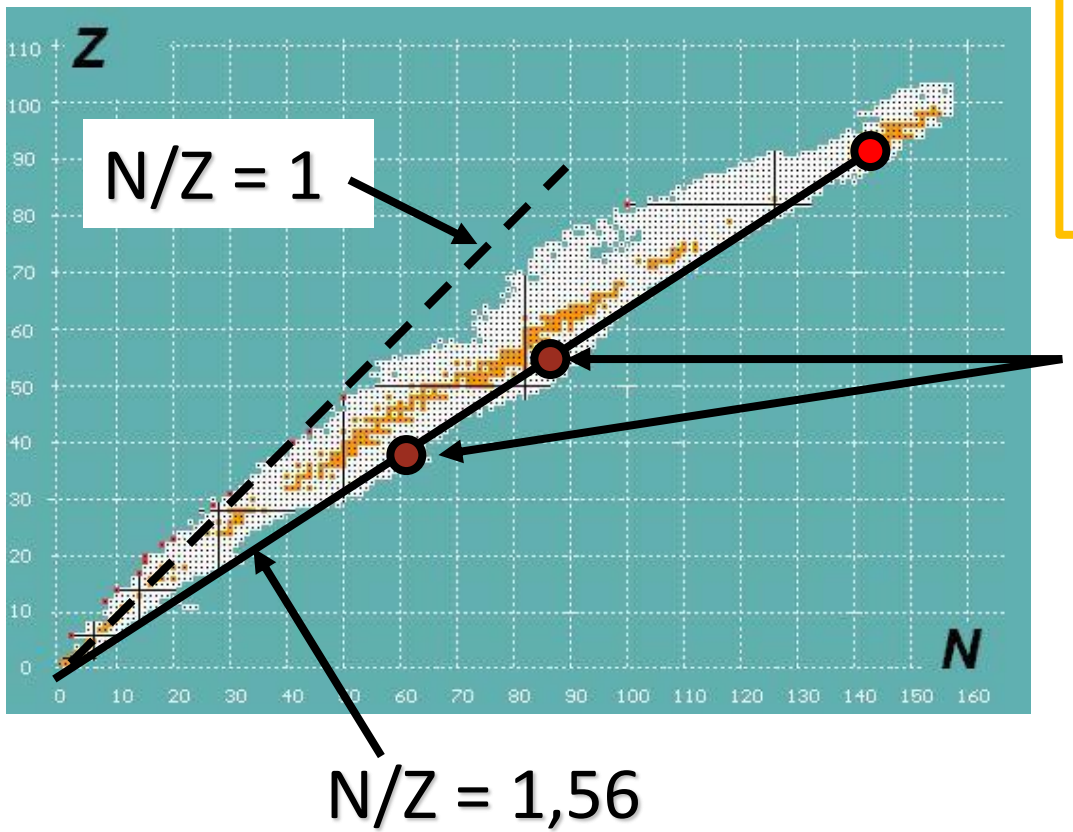


- Starting from the valley of stability with large neutron excess
- Ending up far from the stability line

Neutron-rich nuclei!

Prompt neutron emission **Radioactive β^- decaying products**

Nuclear fission: neutron production



Neutron-rich nuclei!

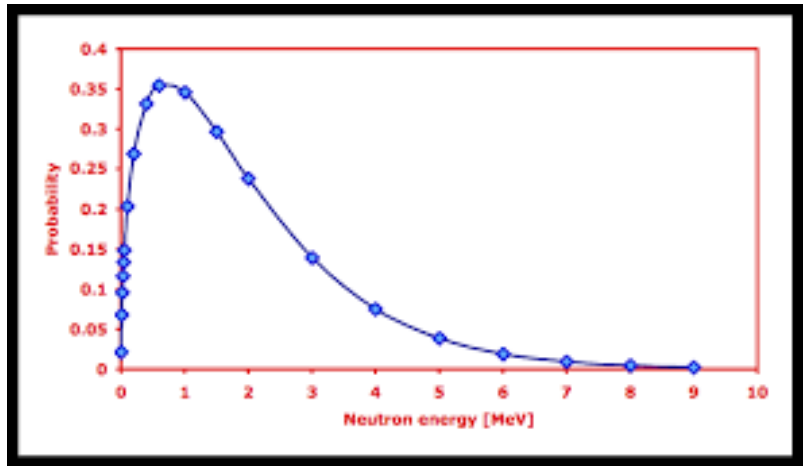
Prompt neutron emission

Radioactive β^- decaying products

Nuclear fission: neutron production

Prompt neutron emission

- Their number follows a Gaussian distribtution
  Average number: 2,4

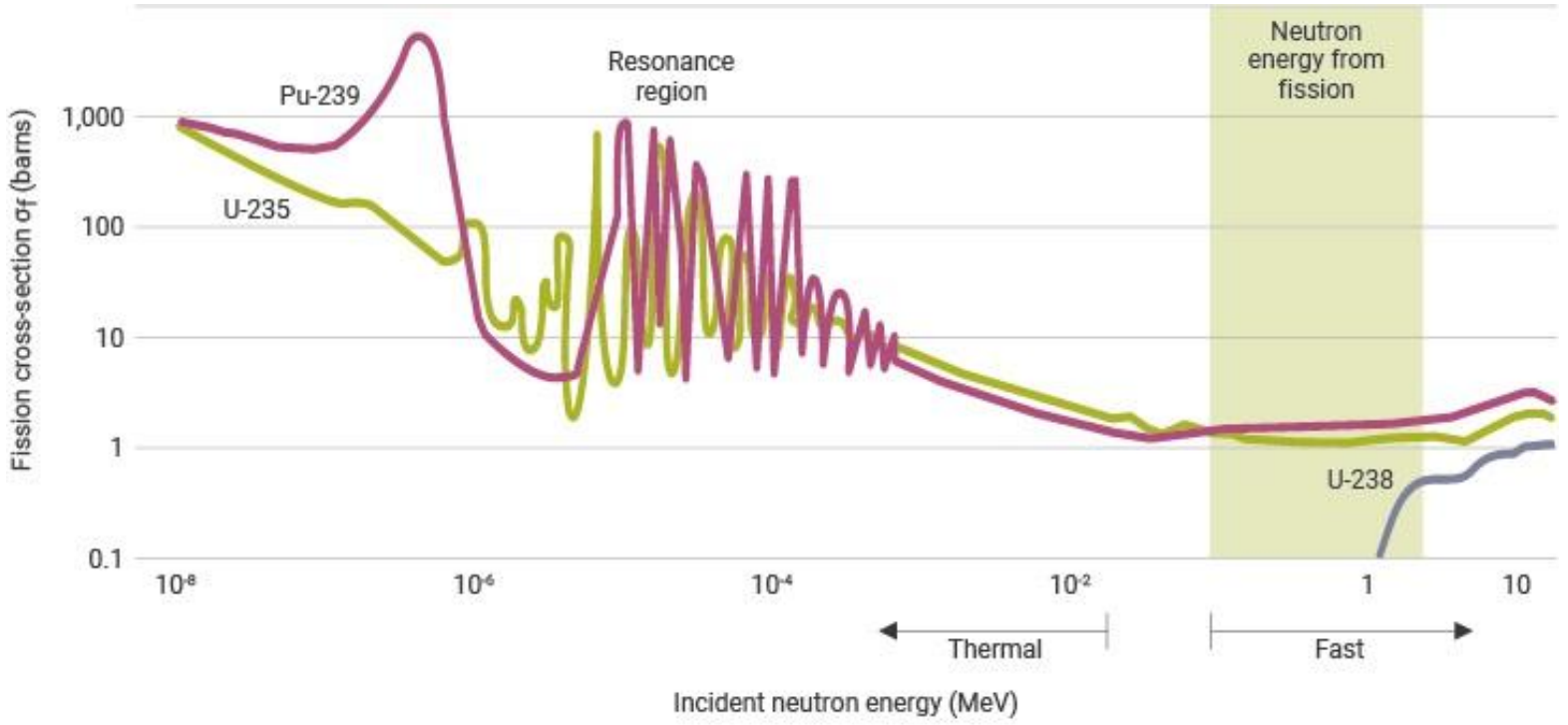


- Energy distribution of prompt neutrons:
  Watt-spectrum

$$N(E) \approx 0,484 \cdot e^{-\frac{E}{E_0}} \cdot \sinh \sqrt{2\frac{E}{E_0}}$$

- Average energy: 2 MeV **high-energy neutrons**!
  - But the fission cross section is small for neutrons in the MeV range, neutrons must be slowed down to be able to induce more fission processes → **moderator** is needed

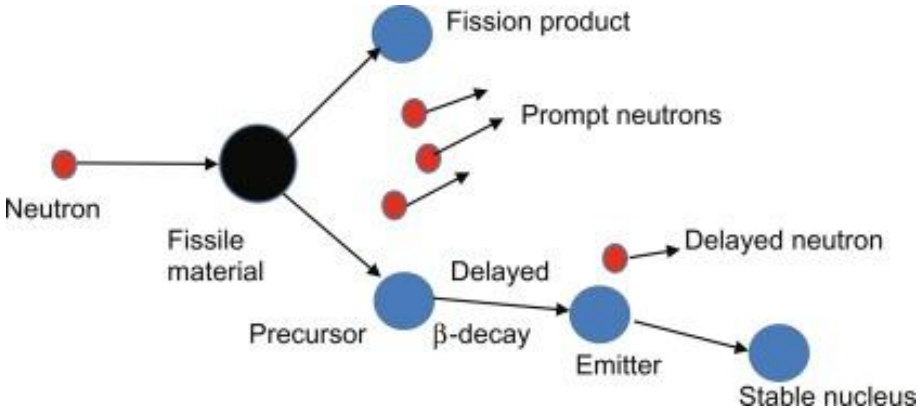
The fission cross section for U and Pu



Nuclear fission: neutron production

Delayed neutrons

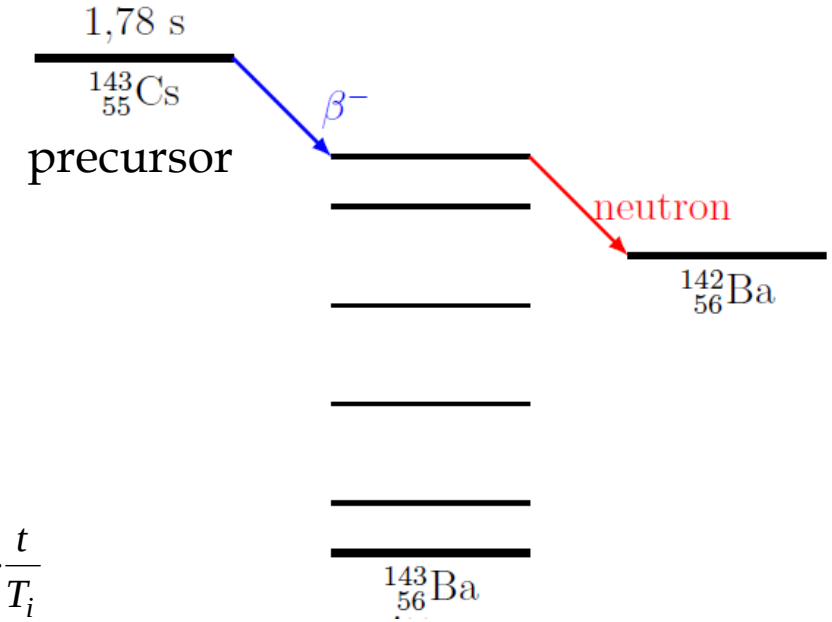
From radioactive β- decaying products



- The delayed neutrons are **emitted with respect to the half-life of the precursor**!

- Temporal appearance:

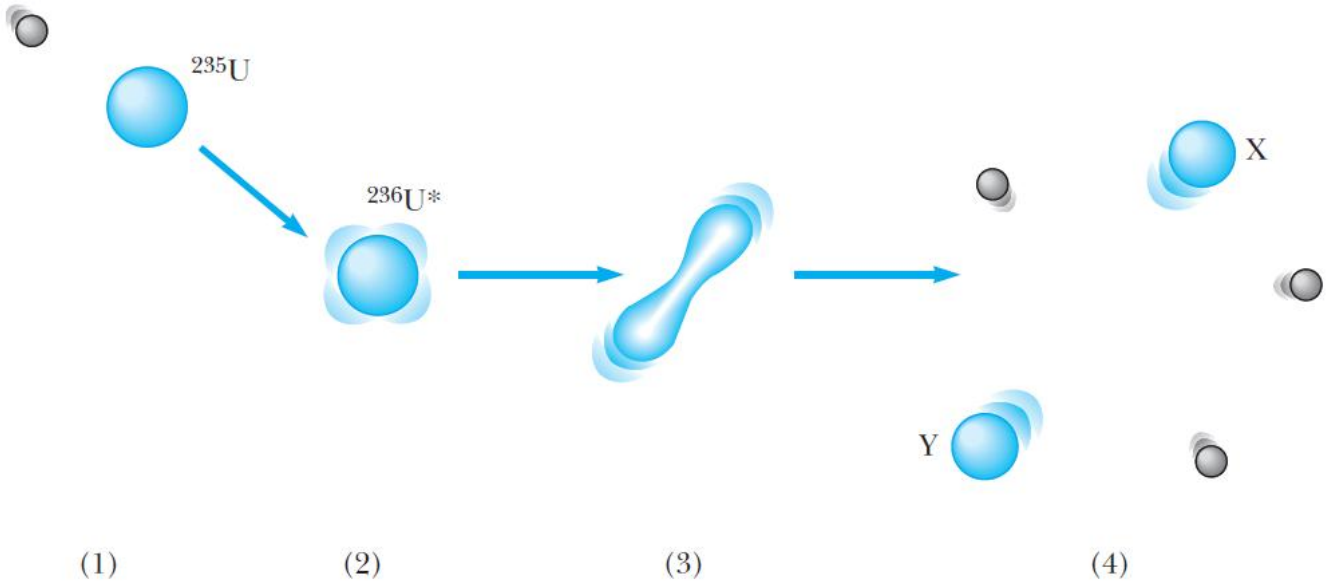
$$N(t) = \sum_{i=1}^{6} \beta_i \cdot e^{-\ln 2 \cdot \frac{t}{T_i}}$$



Nuclear fission: energy balance (U235)

Kinetic energy of fission products	**168 MeV (82,0 %)**
Energy carried away by fission neutrons	5 MeV (2,4 %)
Energy of prompt gamma photons	7 MeV (3,4 %)
Energy carried by beta particles of fission products…	8 MeV (3,9 %)
Gamma radiation of fission products…	7 MeV (3,4 %)
Energy carried by antineutrinos emitted during beta decay of fission products	10 MeV (4,9 %)
SUM	205 MeV (100%)

Nuclear fission: the process



Neutron absorption

Compound nucleus forms at an excited state

Deformation!

Fission into two parts + neutrons

Nuclear fission: fission barrier

$$B = e_V A - e_F \cdot A^{2/3} - e_C \cdot \frac{Z^2}{A^{1/3}} - e_A \cdot \frac{(N-Z)^2}{A} + e_P \cdot \delta \cdot A^{4/3}$$

Small deformation: ε

How much energy needed?

During deformation

- **volume = constant,**
- **asymmetry, pair energy = take as a constant**

- **surface → increases**

$$e_F A^{2/3} \Rightarrow e_F A^{2/3}\left(1 + a_F \cdot \varepsilon\right)$$

- **Coulomb energy → decreases**
  (protons move farther away)

$$e_C \frac{Z^2}{A^{1/3}} \Rightarrow e_C \frac{Z^2}{A^{1/3}}\left(1 - a_C \cdot \varepsilon\right)$$

Nuclear fission: fission barrier

- **surface → increases**
- **Coulomb energy → decreases**
  (protons move farther away)

$$e_F A^{2/3} \Rightarrow e_F A^{2/3}\left(1 + a_F \cdot \varepsilon\right)$$

$$e_C \frac{Z^2}{A^{1/3}} \Rightarrow e_C \frac{Z^2}{A^{1/3}}\left(1 - a_C \cdot \varepsilon\right)$$

$a_F \sim 0{,}025$ **and** $a_C \sim 0{,}012$.

$$\Delta E = \left(0{,}025 \cdot e_F \cdot A^{2/3} - 0{,}012 \cdot e_C \frac{Z^2}{A^{1/3}}\right) \cdot \varepsilon$$

$\Delta E < 0$

Spontaneous fission
Any small deformation is energy-yielding

$\Delta E > 0$

Fission barrier

Nuclear fission: fission barrier

$$\Delta E = \left(0{,}025 \cdot e_F \cdot A^{2/3} - 0{,}012 \cdot e_C \frac{Z^2}{A^{1/3}} \right) \cdot \varepsilon$$

$$\Delta E < 0$$

Spontaneous fission
Any small deformation is energy-yielding

$$\frac{Z^2}{A} \geq \frac{0{,}025 \cdot e_F}{0{,}012 \cdot e_C} \approx 54$$

For heavy nuclei:

$Z/A \sim 0{,}39$

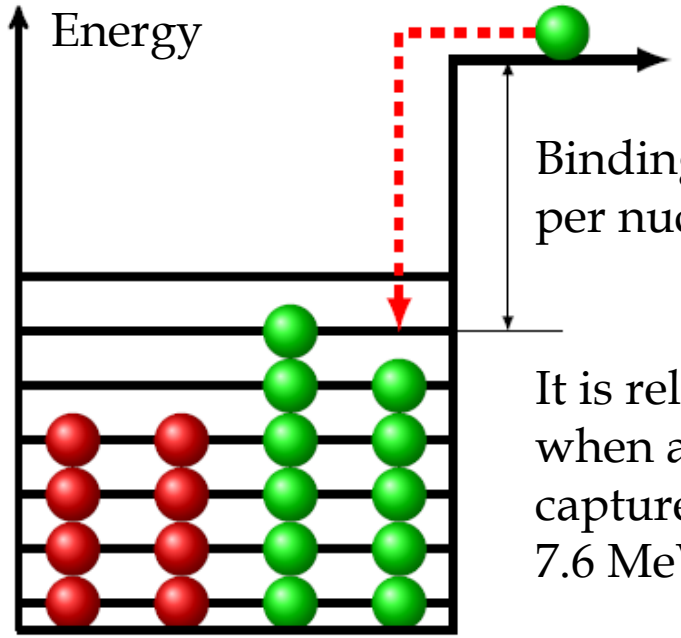
Z>136 nuclei fission spontaneously; no fission barrier

Z>110 small fission barrier, high tunneling probability

Periodic table ends due to spontaneous fission!

Nuclear fission: fission barrier

$$\Delta E = \left(0{,}025 \cdot e_F \cdot A^{2/3} - 0{,}012 \cdot e_C \frac{Z^2}{A^{1/3}} \right) \cdot \varepsilon$$



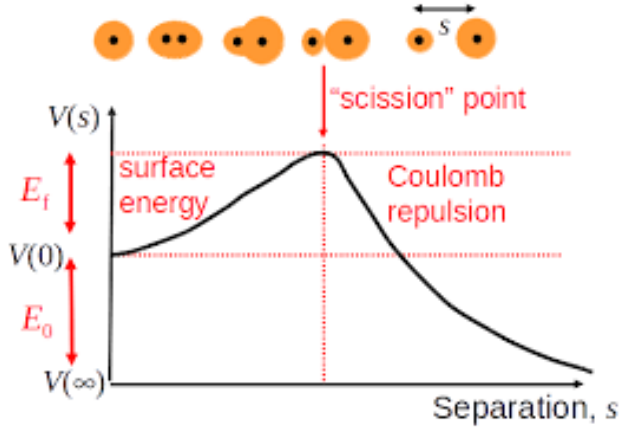
Binding energy per nucleon

It is released when a neutron is captured! (around 7.6 MeV in U236)

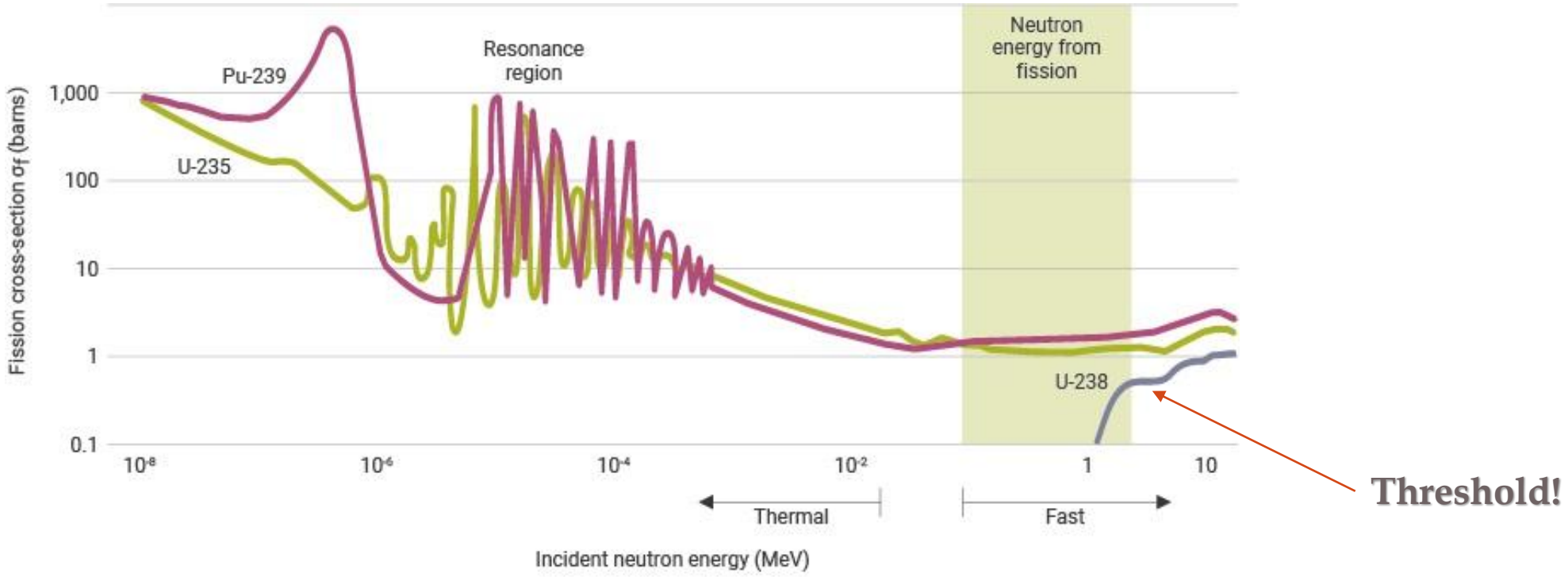
$$\Delta E > 0$$

Fission barrier

7-8 MeV in U!



The fission cross section for U and Pu



Threshold!

Nuclear fission: fission barrier

$$\Delta E = \left(0{,}025 \cdot e_F \cdot A^{2/3} - 0{,}012 \cdot e_C \frac{Z^2}{A^{1/3}} \right) \cdot \varepsilon$$

$$\Delta E > 0$$

Induced fission

Pair energy:

^{235}U + n **produces an even-even nucleus (**^{236}U**)** ⟶ **higher** Q

^{238}U + n **produces an even-odd nucleus (**^{239}U**)** ⟶ **lower** Q

^{235}U + n **can proceed with low-energy neutrons,**

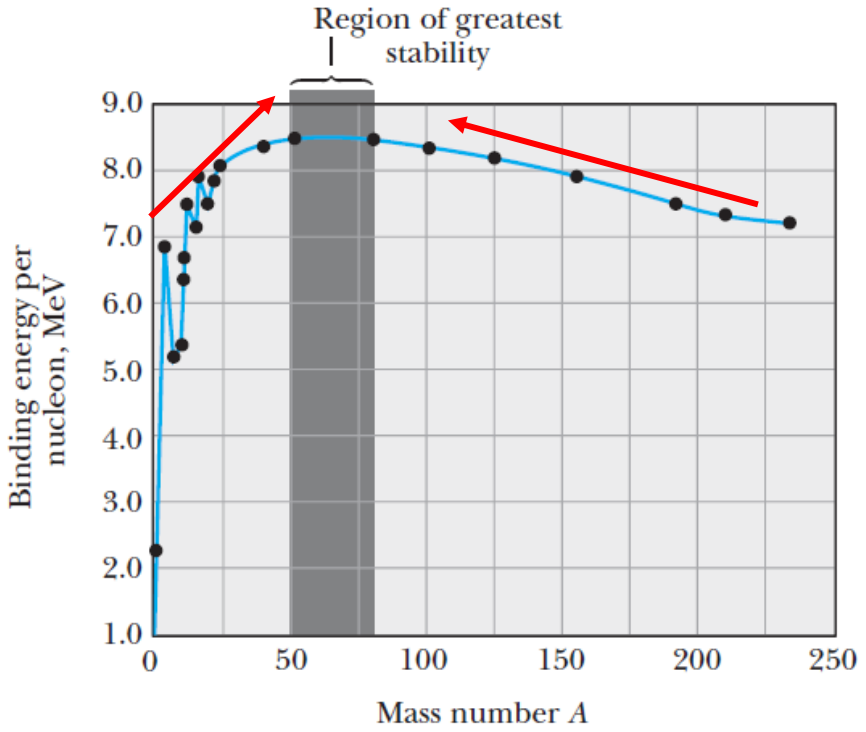
^{238}U +n **has a threshold energy (~0,8 MeV)**

Nuclear fusion reactions

FUSION PROCESSES IN THE EARLY UNIVERSE, FUSION IN THE SUN, FUSION ON EARTH

Nuclear energy release: fusion

- **Fusion**
- Large energy gain /nucleon (2-5 MeV)
- Few nucleons
- Energy gain: **2-18 MeV**



- **Fission**
- Smaller energy gain /nucleon (around 0.8 MeV)
- Many nucleons!
- Larger total energy gain: **200 MeV**

Nucleosynthesis: the beginning

Before the *weak freeze-out* – **thermal equilibrium between p and n (around 1 MeV temperature)**

- ***p/n ratio?***

before the freeze-out: fast weak int. – conversion between p and n maintained

Protons and neutron forming a deuteron – but can only remain **stable** against *photo desintegration* after temperature drops to **0.1 MeV** — **deuteron bottleneck**

$d+\gamma \rightarrow p+n$

B_d=2.2 MeV (small…)
get primordial deuteron by p+n reaction

- ***p/n ratio*** could increase due to neutron beta decay

When **light elements start forming** (p and n locked in the nucleus) – **3 min to 300 000 years**

- p/n stabilizes

- **When stars form primordial free neutrons are extinct due to decay and deuteron formation** (and this deuteron is quickly burned to He (p+d reactions))

Nuclear fusion in stars

Gravitational collapse → **higher temperature -** *estimate the interior temperature of our Sun* – *use the average kinetic energy of the protons*

The core of the star is in a **plasma state (10^7-10^8 K)** → more collapse → **fusion processes**

$$P_{\mathrm{MB}}(E) \propto e^{-m\langle v^2\rangle/2k_{\mathrm{B}}T} = e^{-E/k_{\mathrm{B}}T}$$

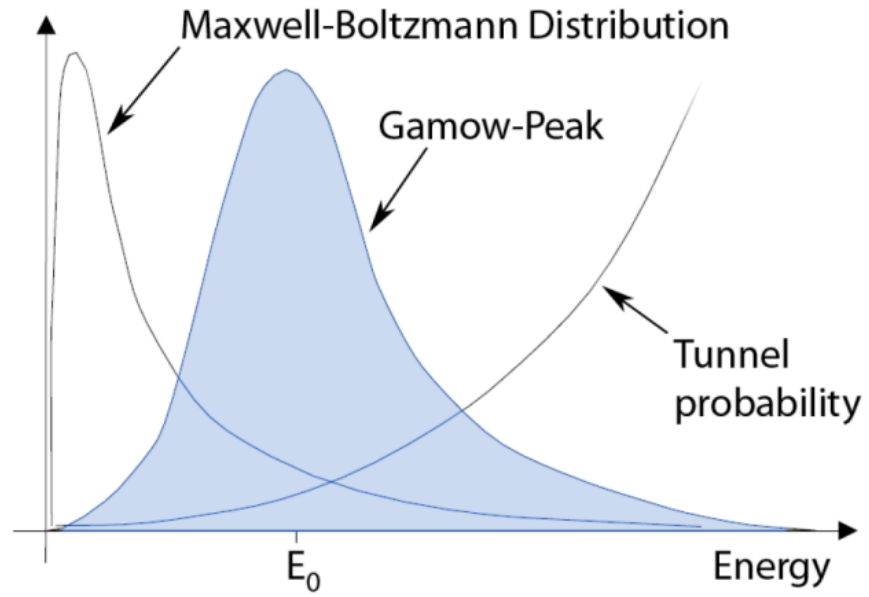
$$P_{\mathrm{G}}(E) = e^{-\sqrt{E_{\mathrm{G}}/E}}$$

Probability that fusion starts

~~$^1_1\mathrm{H} + {}^1_1\mathrm{H} \rightarrow {}^2_2\mathrm{He}$~~

$$^1_1\mathrm{H} + {}^1_1\mathrm{H} \rightarrow {}^2_1\mathrm{H} + e^+ + \nu \qquad (Q = 0,42\ \mathrm{MeV})$$

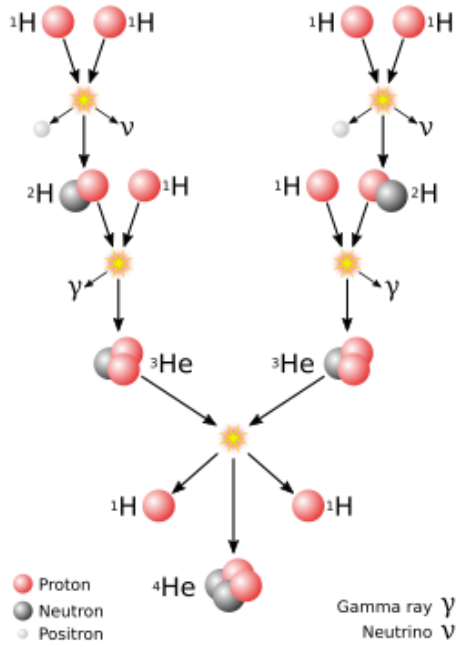
Proceeds via weak interaction!



Nuclear fusion in stars

fusion starts -- how to produce light elements elements? Second step is He!

$${}^1_1\mathrm{H} + {}^1_1\mathrm{H} \rightarrow {}^2_1\mathrm{H} + e^+ + \nu \qquad (Q = 0,42\ \mathrm{MeV})$$



produces most of our Sun's energy

He production

- pp process: 4p → 1 He
- CNO cycle (catalyzed fusion reactions)

- **p-p** starts via weak interaction! Small cross section for weak interaction →slow, but smaller Coulomb barrier
- **CNO** has a bit higher barrier (prop. to $Z_1 * Z_2$), but not initiated by the weak interaction →faster

Created by the triple alpha process ($3\alpha \rightarrow {}^{12}C$)

